SAVEETHA SCHOOL OF ENGINEERING

CSA 15- CLOUD COMPUTING AND BIG DATA ANALYTICS

LIST OF EXPERIMENT 14

14. Demonstrate Infrastructure as a Service (IaaS) by creating a Virtual Machine using a Public Cloud Service Provider (Azure), configure with required memory and CPU

Aim

To demonstrate **Infrastructure as a Service (IaaS)** by creating and configuring a **Virtual Machine (VM)** using **Microsoft Azure** with the required CPU, memory, storage, and networking resources.

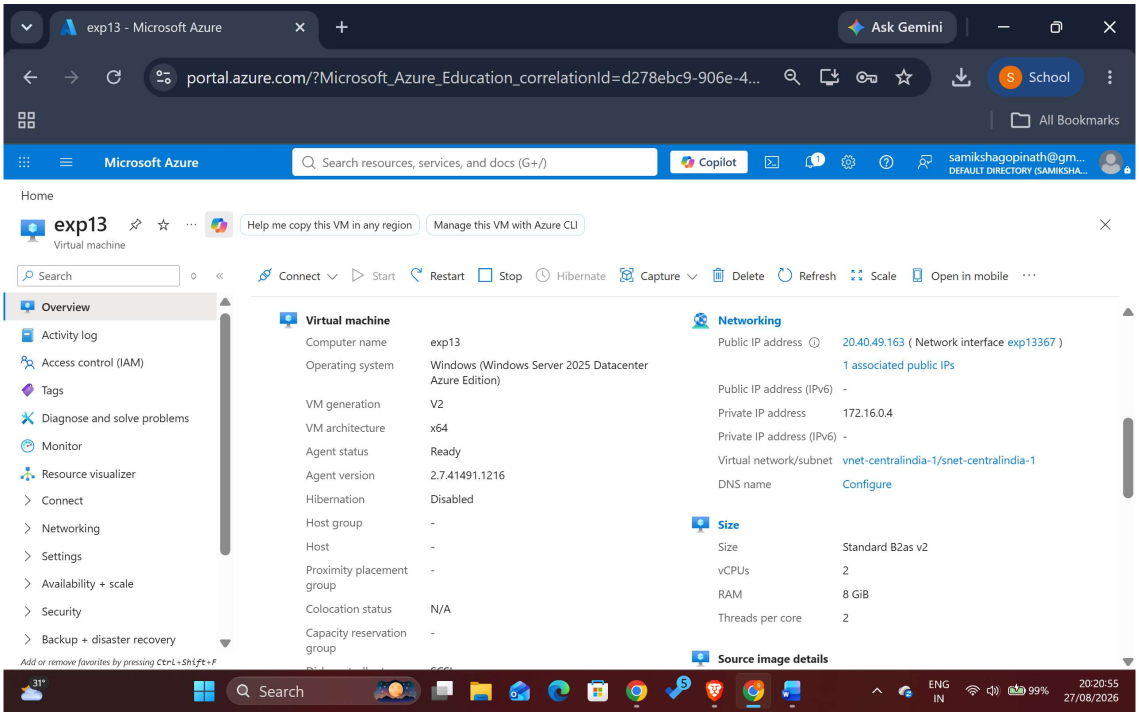
Procedure

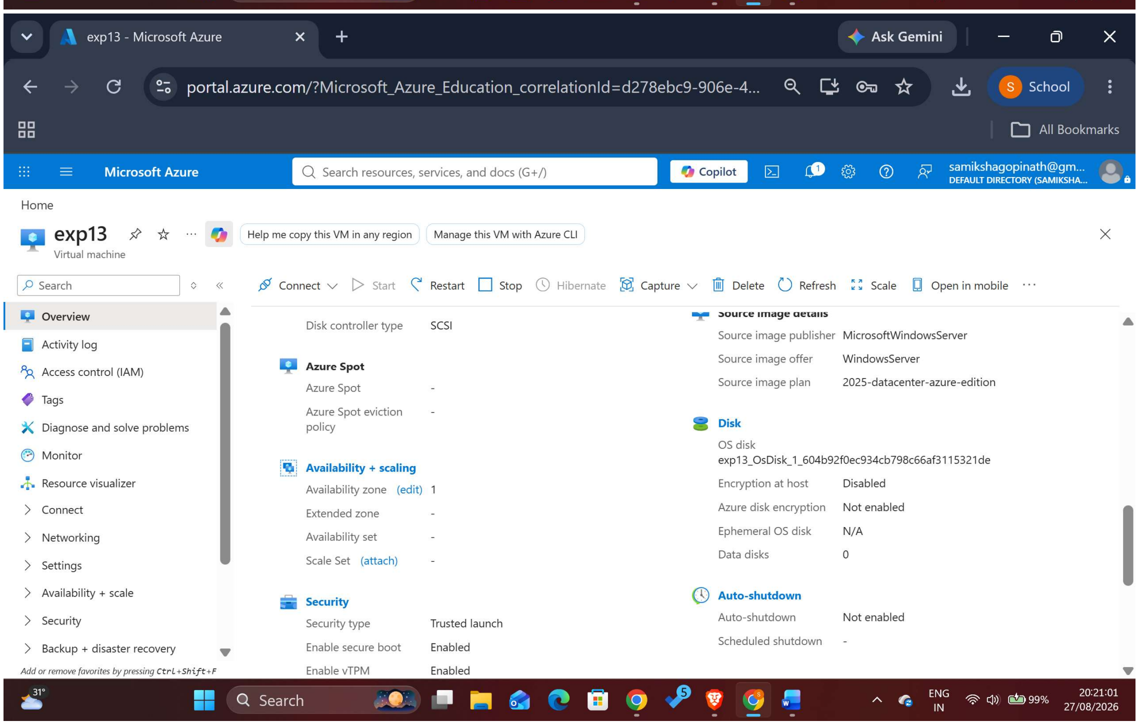
1. Sign in to the **Microsoft Azure Portal**.
2. Select **Virtual Machines** from the Azure services.
3. Click **Create → Azure Virtual Machine**.
4. Select the **Subscription** and create/select a **Resource Group**.
5. Enter a suitable **Virtual Machine Name** and select the required **Region**.
6. Choose the required **Operating System Image**, such as Windows Server or Ubuntu Linux.
7. Under **Size**, select a VM configuration with the required **CPU (vCPUs)** and **Memory (RAM)**.
8. Configure the **Administrator Account** with a username and password or SSH key.
9. Configure the required **Disk/Storage** settings.
10. Configure **Networking** by selecting the Virtual Network, Subnet, Public IP, and Network Security Group.
11. Allow the required inbound port, such as **RDP (3389)** for Windows or **SSH (22)** for Linux.
12. Click **Review + Create** and wait for Azure validation.
13. Click **Create** to deploy the Virtual Machine.
14. After deployment, open the VM and verify that its status is **Running**.
15. Connect to the VM using **Remote Desktop** for Windows or **SSH** for Linux.
16. Verify that the configured CPU, memory, storage, and operating system are available.

Result

A **Virtual Machine was successfully created and configured on Microsoft Azure** with the required CPU, memory, storage, and networking resources. Thus, the **Infrastructure as a Service (IaaS)** model was successfully demonstrated using Azure.

Screenshot:


exp13 - Microsoft Azure
Ask Gemini
portal.azure.com/?Microsoft_Azure_Education_correlationId=d278ebc9-906e-4...
School
All Bookmarks
Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
samikshagopinath@gm...
DEFAULT DIRECTORY (SAMIKSHA...
Home
exp13
Virtual machine
Help me copy this VM in any region
Manage this VM with Azure CLI
Search
Connect
Start
Restart
Stop
Hibernate
Capture
Delete
Refresh
Scale
Open in mobile
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Monitor
Resource visualizer
Connect
Networking
Settings
Availability + scale
Security
Backup + disaster recovery
Add or remove favorites by pressing Ctrl+Shift+F
Virtual machine
Computer name exp13
Operating system Windows (Windows Server 2025 Datacenter Azure Edition)
VM generation V2
VM architecture x64
Agent status Ready
Agent version 2.7.41491.1216
Hibernation Disabled
Host group -
Host -
Proximity placement group -
Colocation status N/A
Capacity reservation group -
Networking
Public IP address 20.40.49.163 (Network interface exp13367)
1 associated public IPs
Public IP address (IPv6) -
Private IP address 172.16.0.4
Private IP address (IPv6) -
Virtual network/subnet vnet-centralindia-1/snet-centralindia-1
DNS name Configure
Size
Size Standard B2as v2
vCPUs 2
RAM 8 GiB
Threads per core 2
Source image details
31°
Search
ENG
IN
99%
20:20:55
27/08/2026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Security
Backup + disaster recovery
Add or remove favorites by pressing Ctrl+Shift+F
Disk controller type SCSI
Azure Spot
Azure Spot -
Azure Spot eviction policy -
Availability + scaling
Availability zone (edit) 1
Extended zone -
Availability set -
Scale Set (attach) -
Security
Security type Trusted launch
Enable secure boot Enabled
Enable vTPM Enabled
Source image details
Source image publisher MicrosoftWindowsServer
Source image offer WindowsServer
Source image plan 2025-datacenter-azure-edition
Disk
OS disk
exp13_OsDisk_1_604b92f0ec934cb798c66af3115321de
Encryption at host Disabled
Azure disk encryption Not enabled
Ephemeral OS disk N/A
Data disks 0
Auto-shutdown
Auto-shutdown Not enabled
Scheduled shutdown -
31°
Search
ENG
IN
99%
20:21:01
27/08/2026